ON THE SURFACE PRESSURE EQUATION ADVECTION DIFFUSION EQUATION ANALOGY

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0.1. **Sign-definiteness of diffusivity.** Let

$$\varphi = \sqrt{\frac{if}{\nu\,\mathrm{Ek}}}, \qquad \mathfrak{A} = \frac{\varphi^{-1}\tanh(\varphi H) - H}{\nu\varphi^2 \mathrm{Ek}}, \tag{1}$$

where $H > 0$, $\nu > 0$, $\mathrm{Ek} > 0$, and $f \neq 0$. Introducing

$$\sigma = \mathrm{sgn}(f), \qquad \alpha = \sqrt{\frac{|f|}{2\nu\,\mathrm{Ek}}}, \qquad s = \alpha H > 0, \tag{2}$$

gives

$$\varphi = \alpha(1 + i\sigma), \qquad \nu\varphi^2 \mathrm{Ek} = if. \tag{3}$$

Using

$$\tanh[s(1+i\sigma)] = \frac{\sinh(2s) + i\sigma\sin(2s)}{\cosh(2s) + \cos(2s)}, \tag{4}$$

we obtain

$$\frac{\tanh(\varphi H)}{\varphi} = \frac{\sinh(2s) + \sin(2s) + i\sigma\,[\sin(2s) - \sinh(2s)]}{2\alpha\,[\cosh(2s) + \cos(2s)]}, \tag{5}$$

$$\mathfrak{A}_r \equiv \mathrm{Re}\,\mathfrak{A} = \frac{\sin(2s) - \sinh(2s)}{2\alpha|f|\,[\cosh(2s) + \cos(2s)]}. \tag{6}$$

For $s > 0$,

$$\cosh(2s) + \cos(2s) > 0 \tag{7}$$

and

$$\sinh(2s) - \sin(2s) > 0, \tag{8}$$

the latter following, for example, from

$$\sinh(2s) - \sin(2s) = \int_0^{2s} (\cosh r - \cos r)\; dr > 0. \tag{9}$$

Consequently,

$$\boxed{\mathfrak{A}_r < 0} \qquad (H > 0,\ \nu > 0,\ \mathrm{Ek} > 0,\ f \neq 0). \tag{10}$$

Thus the positive-definite diffusivity is

$$K \equiv -\mathfrak{A}_r = \frac{\sinh(2s) - \sin(2s)}{2\alpha|f|\,[\cosh(2s) + \cos(2s)]} > 0. \tag{11}$$

The pressure-driven transport is therefore strictly down-gradient:

$$\boldsymbol{U}_{p,\mathrm{diss}} = \mathfrak{A}_r \nabla p_s = -K \nabla p_s. \tag{12}$$

Equivalently, the second-order part of the surface-pressure equation is

$$\nabla \cdot (\mathfrak{A}_r \nabla p_s) = -\nabla \cdot (K \nabla p_s), \qquad K > 0. \tag{13}$$

0.2. **Peclet Number.** The surface-pressure equation may be written as

$$-\nabla \cdot (K \nabla p_s) + \boldsymbol{c} \cdot \nabla p_s = S, \tag{14}$$

where

$$K = -\mathfrak{A}_r > 0, \qquad \boldsymbol{c} = (\partial_y \mathfrak{A}_i, -\partial_x \mathfrak{A}_i) = \nabla^{\perp} \mathfrak{A}_i. \tag{15}$$

A natural local Péclet number based on a horizontal length scale L is

$$\boxed{\mathrm{Pe}_L = \frac{L|\boldsymbol{c}|}{K} = \frac{L|\nabla \mathfrak{A}_i|}{K}}. \tag{16}$$

Consequently,

$$\boxed{0 \le \mathrm{Pe}_L < \infty}. \tag{17}$$

In particular, $\mathrm{Pe}_L = 0$ whenever $\mathfrak{A}_i$ is spatially uniform. This occurs, for example, when both H and f are horizontally uniform. There is no universal finite upper bound because $|\nabla \mathfrak{A}_i|$ may be arbitrarily large.

To examine the coefficient dependence, define

$$\alpha = \sqrt{\frac{|f|}{2\nu\,\mathrm{Ek}}}, \qquad s = \alpha H. \tag{18}$$

The positive diffusivity is

$$K = \frac{\sinh(2s) - \sin(2s)}{2\alpha|f|\,[\cosh(2s) + \cos(2s)]}, \tag{19}$$

and the imaginary part of the transport coefficient is

$$\mathfrak{A}_i = \frac{\mathrm{sgn}(f)}{\alpha|f|}\left[s - \frac{\sinh(2s) + \sin(2s)}{2\,[\cosh(2s) + \cos(2s)]}\right]. \tag{20}$$

If $\mathfrak{A}_i$ varies by an amount comparable to $|\mathfrak{A}_i|$ over the horizontal scale L, then

$$L|\nabla \mathfrak{A}_i| \sim |\mathfrak{A}_i|, \tag{21}$$

and hence

$$\mathrm{Pe}_L \sim \frac{|\mathfrak{A}_i|}{K}. \tag{22}$$

The corresponding coefficient ratio is

$$\frac{|\mathfrak{A}_i|}{K} = \frac{|2s\,[\cosh(2s) + \cos(2s)] - \sinh(2s) - \sin(2s)|}{\sinh(2s) - \sin(2s)}. \tag{23}$$

In the shallow or strongly frictional limit,

$$s \ll 1, \tag{24}$$

Taylor expansion gives

$$\frac{|\mathfrak{A}_i|}{K} = \frac{4}{5}s^2 + O(s^6). \tag{25}$$

Thus,

$$\mathrm{Pe}_L \sim \frac{4}{5}s^2 \ll 1. \tag{26}$$

In the deep or weakly frictional limit,

$$s \gg 1, \tag{27}$$

the coefficient ratio satisfies

$$\frac{|\mathfrak{A}_i|}{K} = 2s - 1 + O\left(e^{-2s}\right). \tag{28}$$

Therefore,

$$\mathrm{Pe}_L \sim 2s = H\sqrt{\frac{2|f|}{\nu\,\mathrm{Ek}}}, \qquad s \gg 1, \tag{29}$$

provided that $\mathfrak{A}_i$ has order-one relative variation over L. Hence the effective Péclet number can become very large when the water depth is large compared with the rotational viscous boundary-layer scale:

$$\boxed{\mathrm{Pe}_L \sim \frac{H}{\sqrt{\nu\,\mathrm{Ek}/(2|f|)}} \gg 1}. \tag{30}$$

Large depth alone, however, does not imply large advection. If H and f are horizontally uniform, then

$$\nabla\mathfrak{A}_i = 0, \qquad \boldsymbol{c} = 0, \qquad \mathrm{Pe}_L = 0, \tag{31}$$

regardless of the value of s.

1. Inferred surface pressure velocity and diffusivity fields: redimensionalization

Let

$$\widehat{q} = \sqrt{\frac{i\widehat{f}}{\widehat{\nu}\,\mathrm{Ek}}}, \qquad \widehat{\mathfrak{A}} = \frac{\widehat{q}^{-1}\tanh(\widehat{q}\widehat{H}) - \widehat{H}}{\widehat{\nu}\,\widehat{q}^{\,2}\mathrm{Ek}}. \tag{32}$$

Define

$$\sigma = \mathrm{sgn}(\widehat{f}), \qquad \widehat{\alpha} = \sqrt{\frac{|\widehat{f}|}{2\widehat{\nu}\,\mathrm{Ek}}}, \qquad s = \widehat{\alpha}\widehat{H}, \qquad D(s) = \cosh(2s) + \cos(2s). \tag{33}$$

Then

$$\widehat{q} = \widehat{\alpha}(1 + i\sigma), \qquad \widehat{\nu}\,\widehat{q}^{\,2}\mathrm{Ek} = i\widehat{f}. \tag{34}$$

Writing

$$\widehat{\mathfrak{A}} = \widehat{\mathfrak{A}}_r + i\widehat{\mathfrak{A}}_i, \tag{35}$$

the real and imaginary parts are

$$\widehat{\mathfrak{A}}_r = \frac{\sin(2s) - \sinh(2s)}{2\widehat{\alpha}\,|\widehat{f}|\,D(s)}, \tag{36a}$$

$$\widehat{\mathfrak{A}}_i = \frac{2sD(s) - \sinh(2s) - \sin(2s)}{2\widehat{\alpha}\,\widehat{f}\,D(s)}. \tag{36b}$$

Equivalently,

$$\boxed{\widehat{\mathfrak{A}}_r = \frac{\sin(2s) - \sinh(2s)}{2\widehat{\alpha}\,|\widehat{f}|\,[\cosh(2s) + \cos(2s)]}} \tag{37}$$

and

$$\boxed{\widehat{\mathfrak{A}}_i = \frac{2s\,[\cosh(2s) + \cos(2s)] - \sinh(2s) - \sin(2s)}{2\widehat{\alpha}\,\widehat{f}\,[\cosh(2s) + \cos(2s)]}}. \tag{38}$$

In particular,

$$\widehat{\mathfrak{A}}_r < 0 \qquad \text{for} \qquad \widehat{H} > 0, \quad \widehat{\nu} > 0, \quad \mathrm{Ek} > 0, \quad \widehat{f} \neq 0. \tag{39}$$

The positive nondimensional diffusivity is therefore

$$\widehat{K} \equiv -\widehat{\mathfrak{A}}_r = \frac{\sinh(2s) - \sin(2s)}{2\widehat{\alpha}\,|\widehat{f}|\,[\cosh(2s) + \cos(2s)]} > 0. \tag{40}$$

Use the dimensional scales

$$x_d = L\widehat{x}, \qquad z_d = H_0\widehat{z}, \qquad H_d = H_0\widehat{H}, \tag{41}$$

$$f_d = f_0\widehat{f}, \qquad \nu_d = \nu_0\widehat{\nu}, \qquad \mathrm{Ek} = \frac{\nu_0}{f_0 H_0^2}, \tag{42}$$

and

$$p_{s,d} = \rho_0 f_0 L U_0\,\widehat{p}_s, \qquad \boldsymbol{\mathcal{U}}_d = H_0 U_0\,\widehat{\boldsymbol{\mathcal{U}}}, \tag{43}$$

where $\boldsymbol{\mathcal{U}}_d$ is the dimensional depth-integrated velocity transport.

If

$$\widehat{\boldsymbol{\mathcal{U}}}_p = \widehat{\mathfrak{A}}\,\widehat{\nabla}\widehat{p}_s, \tag{44}$$

then

$$\boldsymbol{\mathcal{U}}_{p,d} = \mathfrak{A}_d\,\nabla_d p_{s,d}, \qquad \mathfrak{A}_d = \frac{H_0}{\rho_0 f_0}\widehat{\mathfrak{A}}. \tag{45}$$

Define

$$q_d = \sqrt{\frac{i f_d}{\nu_d}}, \tag{46}$$

and introduce

$$\sigma_d = \mathrm{sgn}(f_d), \qquad \alpha_d = \sqrt{\frac{|f_d|}{2\nu_d}}, \qquad s_d = \alpha_d H_d. \tag{47}$$

The dimensional Ekman depth is $1/\alpha_d$, so s_d is the fluid depth divided by the Ekman depth.

The parameter s is invariant under the redimensionalization:

$$s = \widehat{\alpha}\widehat{H} = \alpha_d H_d = s_d. \tag{48}$$

The dimensional complex coefficient is

$$\mathfrak{A}_d = \frac{q_d^{-1}\tanh(q_d H_d) - H_d}{\rho_0 \nu_d q_d^2} = \frac{q_d^{-1}\tanh(q_d H_d) - H_d}{i\rho_0 f_d}. \tag{49}$$

Writing

$$\mathfrak{A}_d = \mathfrak{A}_{r,d} + i\mathfrak{A}_{i,d}, \tag{50}$$

its dimensional real and imaginary parts are

$$\mathfrak{A}_{r,d} = \frac{\sin(2s_d) - \sinh(2s_d)}{2\rho_0\alpha_d\,|f_d|\,[\cosh(2s_d) + \cos(2s_d)]}, \tag{51a}$$

$$\mathfrak{A}_{i,d} = \frac{2s_d\,[\cosh(2s_d) + \cos(2s_d)] - \sinh(2s_d) - \sin(2s_d)}{2\rho_0\alpha_d\,f_d\,[\cosh(2s_d) + \cos(2s_d)]}. \tag{51b}$$

Equivalently,

$$\boxed{\mathfrak{A}_{r,d} = \frac{\sin(2s_d) - \sinh(2s_d)}{2\rho_0\alpha_d\,|f_d|\,[\cosh(2s_d) + \cos(2s_d)]} < 0} \tag{52}$$

and

$$\boxed{\mathfrak{A}_{i,d} = \frac{2s_d\,[\cosh(2s_d) + \cos(2s_d)] - \sinh(2s_d) - \sin(2s_d)}{2\rho_0\alpha_d\,f_d\,[\cosh(2s_d) + \cos(2s_d)]}}. \tag{53}$$

The positive dimensional diffusivity coefficient is

$$K_d \equiv -\mathfrak{A}_{r,d} = \frac{\sinh(2s_d) - \sin(2s_d)}{2\rho_0\alpha_d\,|f_d|\,[\cosh(2s_d) + \cos(2s_d)]} > 0. \tag{54}$$

Thus the pressure-driven dimensional transport is

$$\boldsymbol{\mathcal{U}}_{p,d} = -K_d \nabla_d p_{s,d} + \mathfrak{A}_{i,d}\,\boldsymbol{k} \times \nabla_d p_{s,d}. \tag{55}$$

1.1. **Advecting velocity streamfunction decomposition.** The dimensional advecting streamfunction is the imaginary part of $\mathfrak{A}_d$. It separates naturally into inviscid and frictional parts:

$$\mathfrak{A}_{i,d} = \mathfrak{A}_{i,d}^{(0)} + \mathfrak{A}_{i,d}^{(\mathrm{fr})}, \tag{56}$$

where

$$\boxed{\mathfrak{A}_{i,d}^{(0)} = \frac{H_d}{\rho_0 f_d}} \tag{57}$$

is the classical inviscid result, and

$$\boxed{\mathfrak{A}_{i,d}^{(\mathrm{fr})} = -\frac{\sinh(2s_d) + \sin(2s_d)}{2\rho_0\alpha_d f_d\,[\cosh(2s_d) + \cos(2s_d)]}} \tag{58}$$

is the frictional correction, with

$$\alpha_d = \sqrt{\frac{|f_d|}{2\nu_0}}, \qquad s_d = \alpha_d H_d. \tag{59}$$

Thus,

$$\boxed{\mathfrak{A}_{i,d} = \frac{H_d}{\rho_0 f_d} - \frac{\sinh(2s_d) + \sin(2s_d)}{2\rho_0 \alpha_d f_d \left[\cosh(2s_d) + \cos(2s_d)\right]}}. \tag{60}$$

The corresponding advecting velocity is

$$\boldsymbol{c}_d = \nabla_d^\perp \mathfrak{A}_{i,d} = \boldsymbol{c}_d^{(0)} + \boldsymbol{c}_d^{(\mathrm{fr})}, \tag{61}$$

where

$$\boldsymbol{c}_d^{(0)} = \nabla_d^\perp \left(\frac{H_d}{\rho_0 f_d} \right), \tag{62}$$

$$\boldsymbol{c}_d^{(\mathrm{fr})} = -\nabla_d^\perp \left\{ \frac{\sinh(2s_d) + \sin(2s_d)}{2\rho_0 \alpha_d f_d \left[\cosh(2s_d) + \cos(2s_d)\right]} \right\}. \tag{63}$$

Here

$$\nabla_d^\perp \psi \equiv \left(\partial_{y_d}\psi, -\partial_{x_d}\psi\right). \tag{64}$$

Because the frictional correction to the advecting streamfunction is a function of depth H_d and Coriolis parameter f_d, it can only increase or decrease the flow along f_d/H_d contours (not H_d contours). For shallow depths, friction opposes the tendency to flow around f_d/H_d contours: it retards the flow.